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A 2$^{\text{nd}}$ Order Fully-Passive Noise-Shaping SAR ADC with Embedded Passive Gain

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***Abstract*—An opamp-free solution to implement 2nd order noise shaping in a successive approximation register analog-to-digital converter is presented. This 2nd order fully-passive noise shaping, which has high power efficiency, is realized by charge-redistribution. A gain of 2 is required in this proposal which is realized by a passive method to save power. A prototype chip is fabricated in a 65-nm CMOS process occupying a core area of 0.0129 mm^2. An ENOB of 10.5-bit is achieved at 64-MHz sampling frequency based on an 8-bit CDAC architecture when the OSR is 4. It dissipates 252.9 μW from a 1.0-V supply and achieves a Walden FoM of 10.9 fJ/conv.-step and a Schreier FoM of 169.9 dB.**

***Index Terms*—Fully-passive, noise shaping, SAR ADC**

I. Introduction

A successive approximation register (SAR) analog-to-digital converter (ADC), which is highly composed of digital circuits, is mainly used in moderate resolution and bandwidth applications. Now, the SAR ADC can achieve high power efficiency and very high speed with the help of process scaling. However, the resolution of a SAR ADC is still restrained by the quantization noise, comparator noise and digital-to-analog converter (DAC) noise. To solve these issues, noise shaping (NS) [1], [2] techniques are introduced into the SAR ADC to improve the resolution. Traditionally, the NS is based on an operational amplifier (opamp) [1] which has low power efficiency. Meanwhile, the design of a required opamp is becoming difficult with the process scaling. In order to avoid the drawbacks of an opamp but preserving the noise shaping effect, a fully-passive noise shaping (FPNS) technique [2] is introduced into the SAR ADC. The FPNS technique can achieve high power efficiency and is scaling friendly.

However, for both existing NS SAR ADCs, they can only realize 1st order NS whose noise shaping effect is limited. As a result, the resolution improvement by 1st order NS is not obvious. Moreover, pattern noise [3] may be generated by 1st order NS. In order to overcome the drawbacks of 1st order NS and maintain the advantages of the FPNS technique, a novel 2nd order FPNS SAR ADC is proposed in this paper.

The proposed 2nd order FPNS SAR ADC is realized by charge-redistribution and only does a minor modification to the original 1st order FPNS architecture. Thus, the 2nd order FPNS also maintains the basic architecture and operation principle of a SAR ADC. Then, the merits of a traditional SAR ADC will pass into the 2nd order FPNS SAR ADC. Meanwhile, a gain of 2 is required to realize the proposed 2nd order NS. In order to save power, a passive gain implementation based on capacitive charge pumps [4] is used in this design. The proposed 2nd order FPNS SAR ADC, fabricated in a 65-nm CMOS technology, achieves 10.5-bit ENOB from an 8-bit capacitive DAC (CDAC) at an over-sampling rate (OSR) of 4. The signal bandwidth is 8 MHz and the sampling frequency is 64 MHz. This design dissipates 252.9 μW with a Walden FoM (FoMW) of 10.9 fJ/conv.-step and a Schreier FoM (FoMS) of 169.9 dB when the power supply is 1.0 V.

The remainder of this work is organized as below. The proposed architecture is analyzed in section II. The detailed circuit implementation is described in section III. Experimental results are shown in section IV. A conclusion is drawn in the final section.

II. The Proposed Architecture

A. Zero implementation in a SAR ADC

The zero of noise transfer function (NTF) is realized by adding an additional capacitor (C_2) in a SAR ADC as shown in Fig. 1. The capacitor C_1 plays the function of a CDAC in the SAR ADC. During conversion time (ϕ_C), the top-plates of the capacitors C_1 and C_2 are connected together through a switch ϕ_{NS1}. Thus, the residue (V_{res}) of the SAR ADC is left on the top-plates of the C_1 and C_2 at the same time after conversion. During sampling operation, the capacitors C_1 and C_2 are disconnected with each other. At this moment, the capacitor C_2 still keeps the residue of previous cycle ($V_{res}(N-1)$) while the capacitor C_1 is used to sample the input of current cycle ($X(N)$). Before the conversion operation of current cycle begins (cycle N), charge redistribution is executed between the capacitors C_1 and C_2. After charge redistribution, an addition of the input ($X(N)$) and the residue of previous cycle ($V_{res}(N-1)$) is realized. The total charge in the capacitors is $C_1X(N)+C_2V_{res}(N-1)$. After conversion

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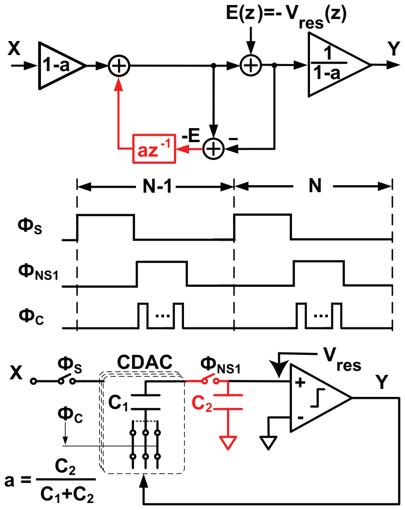

Fig. 1. An architecture of zero implementation in a SAR ADC.

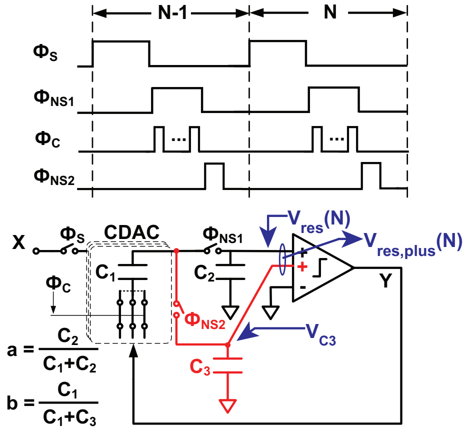

Fig. 2. An architecture of two zeros and one pole implementation in a SAR ADC.

operation at cycle N, according to the charge conservation, the relationship between the input and output is $C_1X(N)+C_2V_{res}(N-1)-C_1Y(N) = (C_1+C_2)V_{res}(N)$. The symbols X and Y represent the input and output of the SAR ADC respectively.

Then, the transfer function can be expressed as

$$Y(z) = X(z) + \frac{1-az^{-1}}{1-a}E(z), \tag{1}$$

where

$$E(z) = -V_{res}(z), a = \frac{C_2}{C_1+C_2}. \tag{2}$$

For this design, the capacitors C_1 and C_2 have the same capacitance ($C_1 = C_2$). Then, the zero is located at 0.5 ($a = 0.5$). A 1st order FPNS SAR ADC by charge redistribution is realized.

B. Two zeros and one pole in a SAR ADC

As shown in Fig. 2, an additional switch (ϕ_{NS2}) and capacitor (C_3) are introduced into the SAR ADC but connecting to an additional positive input of the comparator. The top-plates of the capacitors C_2 and C_3 are connected to two different positive inputs of the comparator.

During the conversion cycle (ϕ_C), the residue is left on the top-plates of the capacitors C_1 and C_2. After conversion and before the next sampling, the capacitors C_1 and C_2 are disconnected with each other. At this moment, the residue is still kept in the capacitor C_1. Then, the capacitors C_1 and C_3 are connected by the switch ϕ_{NS2}. The residue from the capacitor C_1 will be accumulated by the capacitor C_3. The voltage of the capacitor C_3 can be derived as

$$V_{C3}(N) = bV_{res}(N-1) + (1-b)V_{C3}(N-1), \tag{3}$$

where

$$b = \frac{C_1}{C_1+C_3}. \tag{4}$$

The equation (3) can be rewritten in z domain and becomes

$$V_{C3}(z) = \frac{bz^{-1}}{1-(1-b)z^{-1}}V_{res}(z). \tag{5}$$

The input X and output Y only act on the capacitors C_1 and C_2. Therefore, the relationship between the input X and output Y must be calculated according to the charge conservation in the capacitors C_1 and C_2. This is similar to the operation principle shown in Fig. 1. Thus, the relationship between the input X and output Y shown in Fig. 2 still can be written as equation (1).

However, there are two residues left on the top-plates of C_2 and C_3 after conversion for the architecture shown in Fig. 2. The residue in C_3 involves in the decision of the digital output of the comparator which means that the residue in C_3 also decides the resolution of the FPNS SAR ADC. A new variable $V_{res,plus}$, as shown in Fig. 2, is defined to denote the two residues in the capacitors C_2 and C_3. Meanwhile, the addition of the two paths (C_2 and C_3) is through two different positive inputs of the comparator. These two addition path may have different addition weight. The weight of the path C_2 is defined as 1 while the weight of the path C_3 is denoted as κ. Then, the final residue $V_{res,plus}$ can be expressed as

$$V_{res,plus}(z) = \kappa V_{C3}(z) + V_{res}(z). \tag{6}$$

Combine the equations (5) and (6), the final residue becomes

$$V_{res,plus}(z) = \frac{\kappa bz^{-1}V_{res}(z)}{1-(1-b)z^{-1}} + V_{res}(z). \tag{7}$$

According to the equations (1) and (7), the transfer function can be derived as

$$Y = X + \frac{\left(az^{-1}-1\right)\left[1-(1-b)\,z^{-1}\right]}{1+(\kappa b+b-1)\,z^{-1}}\frac{V_{res,plus}(z)}{1-a}. \tag{8}$$

As shown in equation (8), two zeros (a and $(1-b)$) and one pole $(1-\kappa b-b)$, which is 2nd order noise shaping, are realized by the proposed FPNS SAR ADC shown in Fig. 2. When the weight κ is equal to $(1-b)/b$, the pole will disappear and only two zeros left in the equation (8).

If the weight κ is larger than $(1-b)/b$, the transfer function has one negative pole in the NTF, which can further enhance the noise shaping effect. The noise shaping effect of the two zeros and one negative pole is better than that of the two zeros architecture.

However, if the weight κ is smaller than $(1-b)/b$, the transfer function has one positive pole, which will cancel out the effect of one zero. Then, only one zero contributes to the noise shaping effect. The noise shaping effect will be weakened by the positive pole. Therefore, it is better to make sure that the weight κ is greater than or equal to $(1-b)/b$.

For this design, the weight κ is 2 and the parameter b is 0.4. As mentioned above, the parameter a is 0.5. Then, the transfer function can be written as

$$Y = X + \frac{\left(0.5z^{-1}-1\right)\left[1-0.6z^{-1}\right]}{1+0.2z^{-1}}\frac{V_{res,plus}(z)}{0.5}. \tag{9}$$

III. Circuit Implementation

A. The operation principle of a passive gain

The weight of the path C_3 also can be seen as the amplification of the voltage on the top-plate of the capacitor C_3. It is easy to realize a passive gain of 2 by capacitive charge pumps [4].

As shown in Fig. 3, the passive gain is realized by two operation phases. During the sampling phase (ϕ_{sa}), the input is sampled onto the top-plates of the two capacitors. Then, each capacitor has a voltage of V_{in} after sampling. During an addition phase (ϕ_{ad}) after sampling, the two capacitors are stacking in serial. The output voltage becomes the addition of the two input voltages which means that the output voltage is $2V_{in}$. As a result, a gain of 2 is realized by this passive implementation.

B. Circuits information

The proposed schematic is shown in Fig. 4. A differential scheme is implemented and a single-ended schematic is shown for simplicity.

The capacitor C_3, which is used to realize a passive gain, is divided into two equal capacitors. The voltages in the C_3 is around the common-mode voltage. Meanwhile, the settling time is given large enough for the charge sharing

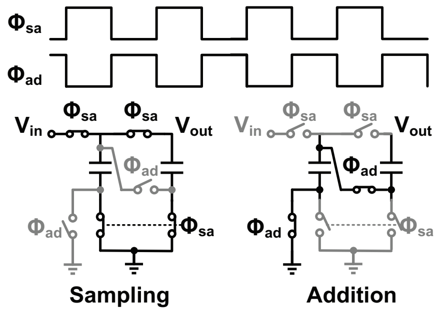

Fig. 3. A demonstration of passive gain implementation.

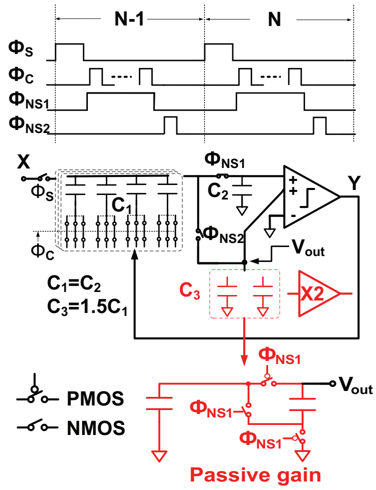

Fig. 4. The proposed 2nd order FPNS SAR ADC schematic.

in the passive gain. Therefore, the switches used in the passive gain architecture are realized by both NMOS and PMOS. Then, only one phase clock, ϕ_{NS1} in this design, is used to control the switches in the passive gain architecture as shown in Fig. 4. When the voltage of ϕ_{NS1} is low and the ϕ_{NS2} is high, the charge sharing between the capacitors C_1 and C_3 is executed. When the ϕ_{NS1} is high, the voltage in C_3 is amplified by 2. The capacitors C_1 and C_2 have the same capacitance while the capacitor C_3 is equal to $1.5C_1$.

The comparator has four inputs and is realized by dynamic architecture for the consideration of low power. Its architecture is shown in Fig. 5.

IV. Measurement Results

The proposed 2nd order FPNS SAR ADC is fabricated in a 65-nm CMOS process and occupies a core area of 0.0129 mm^2. The die photograph is shown in Fig. 6. With 64 MS/s and 1.0-V supply, Fig. 7 shows the measured spectrum with 2nd order noise shaping, giving an SNDR

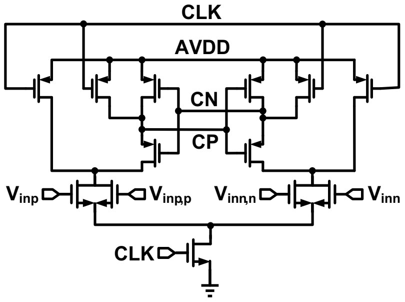

Fig. 5. A four inputs dynamic comparator architecture.

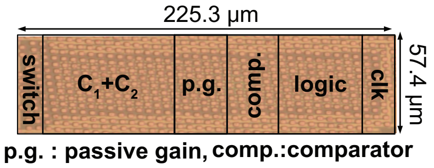

Fig. 6. Chip photograph.

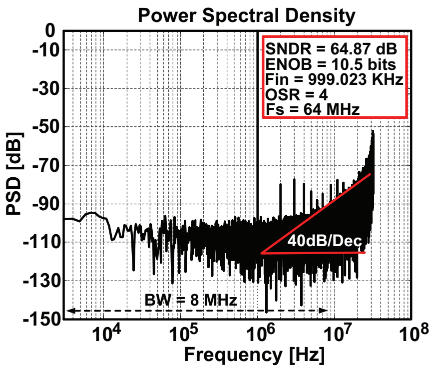

Fig. 7. The measured PSD.

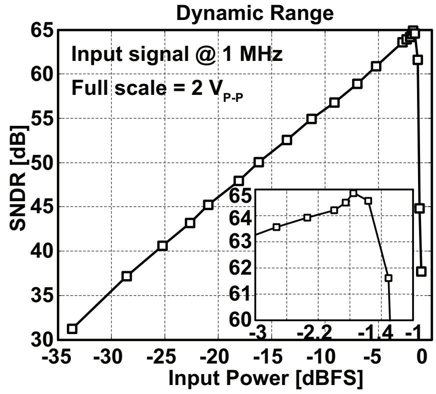

Fig. 8. The SNDR vs. input power.

of 64.9 dB (10.5-bit ENOB) and an SFDR of 77.5 dB. The power consumption is 252.9 μW where analog power is 74.3 μW and digital power is 178.6 μW. The power consumption can be further reduced with the process scaling which means that a better performance can be expected in the near future. A FoMW of 10.9 fJ/conv.-step and a FoMS of 169.9 dB are achieved by the proposed 2nd order FPNS SAR ADC. The relationship between the SNDR and input power is shown in Fig. 8.

The performance comparison with previous NS SAR ADCs is shown in Table I. All three NS SAR ADCs in the table are based on an 8-bit CDAC architecture. This work achieves the best resolution improvement. Meanwhile, this work achieves the best FoMW and FoMS.

TABLE I
PERFORMANCE COMPARISON.

	[1]	[2]	This work
Architecture	NS-SAR	FPNS-SAR	FPNS-SAR
Use opamp?	Yes	No	No
Noise shaping order	1	1	2
Technology (nm)	65	65	65
Supply voltage (V)	1.2	0.8	1
Core area (mm^2)	0.0323	0.0123	0.0129
Power (μW)	806	120.7	A:74.3 D:178.6
OSR	4	4	4
Bandwidth (MHz)	11	6.25	8
ENOB (bits)	10	9.35	**10.5**
FoMW (fJ/conv.-step)	35.8	14.8	**10.9**
FoMS (dB)	163.4	165.1	**169.9**

V. CONCLUSION

An opamp-free 2nd order FPNS SAR ADC, which has high power efficiency, is realized in this design. The noise shaping effect and performance are improved when compared with traditional NS SAR ADCs.

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